Introduction to Mobile Computing

The rapidly expanding technology of cellular communication, wireless LANs, and satellite services will make information accessible anywhere and at any time. Regardless of size, most mobile computers will be equipped with a wireless connection to the fixed part of the network, and, perhaps, to other mobile computers. The resulting computing environment, which is often referred to as ***mobile or nomadic computing***, no longer requires users to maintain a fixed and universally known position in the network and enables almost unrestricted mobility. Mobility and portability will create an entire new class of applications and, possibly, new massive markets combining personal computing and consumer electronics.

Mobile Computing is an umbrella term used to describe technologies that enable people to access network services anyplace, anytime, and anywhere.

(i). Wireless Computing

Refers to ***computing systems that are connected to their working environments via wireless links*** (such as radio frequency (RF) or infrared (IR)). Wireless computing is based on a more conventional extension of networked computing.

This term generally refers to computing devices participating in a wireless LAN, with gateways to more traditional wired networks.

The key concept here is the ability to participate in a workgroup – a collection of computing devices and servers for sharing data and information. Wireless email is a form of wireless computing.

(ii). Ubiquitous Computing

This paradigm envisions an ***environment in which computing devices are so inexpensive and readily available that there are hundreds and even thousands in an office***.

With such a large number of devices, any attempt to interconnect them with traditional wired networks would be physically impossible; thus wireless communications would be a necessary part of the infrastructure.

Ubiquitous devices individually demand relatively modest bandwidth (e.g 64 kbps or less), although the aggregate within the volume of a typical office may be high.

Within a few years, it is expected that many of the world's fridges, heart monitors, bus ticket dispensers, burglar alarms, and electricity meters will be sending messages to each other. This ubiquitous computing vision – of spontaneous interaction between the digital devices that serve us – could bring great impact on society

Indeed, a common view of the Internet divides it into three waves: originally, mainframes and terminals; until yesterday, PCs, browsers and a GUI; starting tomorrow, wirelessly networked processors embedded in everyday objects. (It is expected that currently, there could be more mobile phones connected to the Internet than computers.)

(iii). Nomadic Computing

Nomadic computing refers to the ***ability to compute as the user relocates from one support environment to another***. This is the wireless equivalent of permitting a non-local user to use a local machine to make remote connections to the user's home system.

In this model, it is expected that individual organizations will construct their own wireless infrastructures, linked across organizations by wired internetworks. However, a user would like to be able to use his or her device even within a foreign organization's wireless infrastructure.

System support for nomadic computing must address the issues of trust, security and privacy as wireless computing devices move between such organizational structures. The host infrastructure must retain security against the nomad, while the operations of the nomad must be kept private from the owners of the host infrastructure.

(iv). Decoupled Computing

Decoupled computing refers to ***the ability to compute when detached from the existing computing and communications infrastructure***.

If the wireless computing devices are full-function computers in their own right – such as modern notebook computers – the decoupled computing makes it possible to perform some operations (like file access from servers) in a transparent manner even when disconnected from the file server.

Using techniques like file caching in the wireless device, it is possible for the user to update the file even when disconnected. When he/she re-establishes connection, the cached files are automatically re-synchronized with the copies on the server.

Decoupled computing is an important element of the more general wireless computing spectrum, because the support infrastructure for wireless will not be universally available for many years to come.

Natural Evolution of Computing

Wireless information systems represent the next logical step in the natural evolution of computing systems. Over time, the user has been freed from the need to be collocated in time and space with the computing system he/she uses – as shown in the figure below.

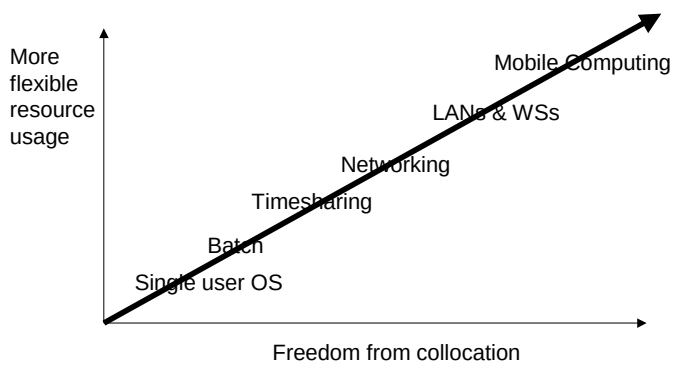


Fig. 1. The Evolution of Computing Systems

Early machines required the user to be physically located at the machine to use it. As they ran a single-user operating system, the computing resource could not be effectively used by more than one user at a time. As usage restrictions are reduced, multi-user collaborative activities are better enabled.

Over time, the evolution of computing has reduced the tight coupling between the user and his computing resource and environment.

Batch processing freed the user in time: the user no longer needed to wait until the computer became free to submit work. But this still required the user to be located with the computer to use it.

Time sharing allowed the computer resource to be simultaneously shared among many users, while adding an element of physical dispersion from the actual machine.

Conventional networking of distributed timesharing systems extended this dispersion across a geographically wider area, but this still limited the ease with which the interconnected computing resources could be shared among users.

A communication device can exhibit any one of the following characteristics:

- **Fixed and wired**: This configuration describes the typical desktop computer in an office. Neither weight nor power consumption of the devices allow for mobile usage. The devices use fixed networks for performance reasons.
- **Mobile and wired**: Many of today's laptops fall into this category; useds carry the laptop from one hotel to the next, reconnecting to the network via the telephone network and a modem.
- **Fixed and wireless**: This mode is used for installing networks, e.g., in historical buildings to avoid damage by installing wires, or at trade shows to ensure fast network setup.
- **Mobile and wireless**: This is the most interesting case. No cable restricts the user, who can roam between different wireless networks.

Some Technical Issues of Wireless Computing (Research Challenges)

Various levels of a wireless information system exist and present their own forms of research challenges; these levels are discussed:

- *Hand-held terminal level; Wireless link level; The base Station level; Network backbone level;*

(i). The Terminal

Mobile terminals are microelectronics assemblies that are attached to sophisticated network-based systems. Key issues to be addressed in building these light-weight wireless devices include:

- ***Display technology*** – colour displays (though preferred) require greater transistor densities and thus are more costly.

- ***Advanced packaging for high circuit density in a small volume*** – packaging remains an important underlying technology as terminal sizes decrease while the embedded processing capabilities increase, especially in support of greater intelligence of the terminal to sense its environment.

- ***Embedded software*** – this implements the various control functions of the mobile terminal (display control, data input and output, and the various control functions layered on top of the

wireless link). The embedded software should provide support for handwriting and speech recognition.

- ***Small, lightweight design*** – this is another key aspect of the mobile terminal; the device must be small and lightweight, as input may be from a pen or speech. A full keyboard would result in devices not much smaller than today's notebook computers.

- ***Battery Technology*** – a variety of improvements in battery technology are proposed (nickel cadmium, nickel, nickel hydride, etc), but improvements of only about 20% in battery lifetime are expected over the next two decades. The best opportunity for extending battery life is not in a new battery technology but reducing the power requirements of the mobile terminal. This can be accomplished through a consistent strategy for lower-power circuit and system design.

  For example, algorithms for image compression/decompression can be used that reduce the computational demands at the expense of more storage in the terminal

(ii). The Wireless Link

The critical issue in the engineering of the wireless link is the efficient encoding of the signal to minimize the bit-error rate in the face of interference and fading. Design considerations include: efficient RF link design, efficient system design, improved frequency reuse, etc.

While existing radio transmission systems use frequency division multiple access (FDMA) or time-division multiple access (TDMA), the general trend appears to be in the direction of code-division multiple access (CDMA), also known as *spread spectrum*.

Spread spectrum techniques spread the communications waveform across a wide range of frequencies – selected to allow multiple waveforms to be superimposed to limit interference.

Spread spectrum has many advantages:

- *significant tolerance to interference (thus permitting higher user capacity);*

- *The ability to reuse frequencies in adjacent cells by simply assigning different spreading codes.*

(iii). The Base Station

Cellular telephone systems critically depend on the supporting infrastructure of *base stations* and *mobile telephone switching offices* (***MTSOs***). Communications need to be re-established rapidly, especially in those situations where the existing wired infrastructure does not exist or has been damaged (as in the case of disasters).

Today, the setup time for base stations is long, requiring significant efforts to set up a cell and prepare the site. Careful planning and positioning of the microwave radios used to link the base stations is necessary.

It is thus necessary for a strategy to develop "instant infrastructures" to support wireless communications. The instant infrastructure could comprise of semi-fixed but rapidly deployable infrastructure.

(iv). Networking

The network design issues include *protocols* and *algorithms* for mobile hosts and multimedia communications. These include:

- Network addressing schemes for network nodes that move;
- Support for real-time packet delivery for continuous media;
- Network scalability (in terms of numbers of users, bandwidth, and distance).

The first issue in wireless networking is *how to access the wireless media*. A standard collision detection scheme can be used (as in Ethernet, CSMA/CD):

- *This requires the sending terminal to be able to hear all other terminals that may attempt to communicate with the base station at the same time*.

- *A terminal on one side of a cell may not be able to detect that it is colliding with a terminal at the opposite end of the cell.*

A proposed solution is the MACA (*media access collision avoidance*) media access protocol. According to this protocol, clients enter into a 4-way handshake (RTS-CTS-DATA-ACK) with an access point:

- Client sends Request-to-Send (RTS) packet to request use of the medium;
- If the medium is free, access point sends the Clear-to-Send (CTS) packet to the client;

Client sends DATA to the receiving client

APPLICATIONS OF MOBILE COMPUTING

In many fields of work, the ability to keep on the move is vital in order to utilize time efficiently. The importance of Mobile Computers has been highlighted in many fields of which a few are described below:

a. **Vehicles:** Music, news, road conditions, weather reports, and other broadcast information are received via digital audio broadcasting (DAB) with 1.5 Mbit/s. For personal communication, a universal mobile telecommunications system (UMTS) phone might be available offering voice and data connectivity with 384 kbit/s. The current position of the car is determined via the global positioning system (GPS). Cars driving in the same area build a local ad-hoc network for the fast exchange of information in emergency situations or to help each other keep a safe distance. In case of an accident, not only will the airbag be triggered, but the police and ambulance service will be informed via an emergency call to a service provider. Buses, trucks, and trains are already transmitting maintenance and logistic information to their home base, which helps to improve organization (fleet management), and saves time and money.

b. **Emergencies**: An ambulance with a high-quality wireless connection to a hospital can carry vital information about injured persons to the hospital from the scene of the accident. All the necessary steps for this particular type of accident can be prepared and specialists can be consulted for an early diagnosis. Wireless networks are the only means of communication in the case of natural disasters such as hurricanes or earthquakes. In the worst cases, only decentralized, wireless ad-hoc networks survive.

c. **Business**: Managers can use mobile computers say, critical presentations to major customers. They can access the latest market share information. At a small recess, they can revise the presentation to take advantage of this information. They can communicate with the office about possible new offers and call meetings for discussing responds to the new proposals. Therefore, mobile computers can leverage competitive advantages. A travelling salesman today needs instant access to the company's database: to ensure that files on his or her laptop reflect the current situation, to enable the company to keep track of all activities of their travelling employees, to keep databases consistent etc.With wireless access, the laptop can be turned into a true mobile office, but efficient and powerful synchronization mechanisms are needed to ensure data consistency.

d. **Credit Card Verification**: At Point of Sale (POS) terminals in shops and supermarkets, when customers use credit cards for transactions, the intercommunication required between the bank central computer and the POS terminal, in order to effect verification of the card usage, can take place quickly and securely over cellular channels using a mobile computer unit. This can speed up the transaction process and relieve congestion at the POS terminals.

e. **Replacement of Wired Networks**: wireless networks can also be used to replace wired networks, e.g., remote sensors, for tradeshows, or in historic buildings. Due to economic reasons, it is often impossible to wire remote sensors for weather forecasts, earthquake detection, or to provide environmental information. Wireless connections, e.g., via satellite, can help in this situation. Other examples for wireless networks are computers, sensors, or information displays in historical buildings, where excess cabling may destroy valuable walls or floors.

f. **Infotainment**: wireless networks can provide up-to-date information at any appropriate location. The travel guide might tell you something about the history of a building (knowing via GPS, contact to a local base station, or triangulation where you are) downloading information about a concert in the building at the same evening via a local wireless network. Another growing field of wireless network applications lies in entertainment and games to enable, e.g., ad-hoc gaming networks as soon as people meet to play together.

Mobile Computing Devices

Mobile Computing Devices

Some of the mobile computing devices have already been mentioned in order to describe the different wireless data transfer options. Following is a closer look at the range of products and functions. Although there are many products on the market, handheld devices fall into several categories, each with specific capabilities and options.

Device Type	Product Features
Web Phone : Cellular phone with Internet access	➢ Internet browser functions ➢ Limited email, calendar, appointment scheduling and directories.
PDA Phone : Combination of web phone with PDA personality	➢ Internet browser functions. ➢ Email, calendar, appointment scheduling and directories.
Personal Digital Assistant(PDA) or Pocket PC Handheld computerized information organizer; examples include Palm Pilot, Handspring Visor, Compaq iPaq	➢ Email, calendar, appointment scheduling, and directories. ➢ Some desktop application functions. E.g., Word and excel. ➢ Pen-based system for data entry ➢ Bar coding.
Handheld PC : Small hand size personal computer with a keyboard	➢ More powerful than a PDA device. ➢ Some desktop application functions. E.g., Word and excel. ➢ Keyboard for data entry. ➢ Voice reorganization and recording option ➢ Bar coding
Tablet/Laptop Tablet is a flat paneled. Laptop is also known as PC notebook	➢ All desktop functionality ➢ Tablets use pen or touch screen technology

Types of wireless Communication Technologies

Satellite_communication

Satellite_communication is one of the wireless technologies, which is widely spread all over the world allowing users to stay connected virtually anywhere on the Earth. The Satellites used in this mode of communication, communicate directly with the orbiting satellites via radio signals. Portable satellite

phones and modems have more powerful broadcasting abilities than the cellular devices as they have high range, apart from being more expensive in terms of cost, than their counterparts. For example, for outfitting a ship through satellite communication, a traditional communication system is linked to a single satellite, which allows multiple users to share the same broadcast equipment.

Bluetooth Technology

Bluetooth technology allows you to connect a variety of different electronic devices wirelessly to a system for the transfer and sharing of data and this is the main function of Bluetooth. Cell phones are connected to hands-free earpieces, wireless keyboard, mouse and mike to laptops with the help of Bluetooth as it transmits information from one device to other device. Bluetooth technology has many functions, and it is used most commonly in wireless communications' market.

Bluetooth Technology

Features

- Bluetooth technology uses radio waves to communicate between devices. Most of these radio waves have a range of 15-50 feet.
- According to the official Bluetooth website, Bluetooth uses a low-power signal with a maximum range of 50 feet with sufficient speed to enable transmission of data.
- The pairing process identifies and connects any two devices to each other. It also prevents interference from other non-paired Bluetooth devices in the area.
- It uses maximum power only when it is required, thus preserving battery life.

ZigBee

ZigBee is a wireless communication standard designed to address the unique needs of low-power, low-cost wireless sensor, and control networks. ZigBee can be used almost anywhere, as it is easy to implement and requires little power to operate. Zigbee has been developed looking into the needs of the communication of data with a simple structure like the data from the sensors.

Zigbee Technology

Features

- ZigBee devices are designed for low-power consumption.
- ZigBee is used in Commercial Applications like sensing and monitoring applications.
- ZigBee uses very low power and extremely long device battery life.
- ZigBee gives flexibility to do more with the reliable wireless performance and battery operation.

Microwave

- Microwave is an effective type of wireless data transmission that transfers information using two separate methods. One method which is used to transmit data through the wireless media of a microwave is the satellite method that transmits information via a satellite that orbits 22,300 miles above the Earth. Stations on the ground send and receive data signals to and from the satellite with a frequency ranging from 11 GHz to 14 GHz and with a transmission speed of 1 Mbps to 10 Mbps. Another method is a terrestrial method, in which two microwave towers with a clear line of sight between them are used ensuring no obstacles to disrupt that line of sight. For the purpose of privacy, it is used often. The frequency of data transmission for terrestrial systems is typically 4 GHz to 6 GHz or 21 GHz to 23 GHz, and the speed is usually 1 megabit per second (Mbps) to 10 Mbps.

Infrared (IR)

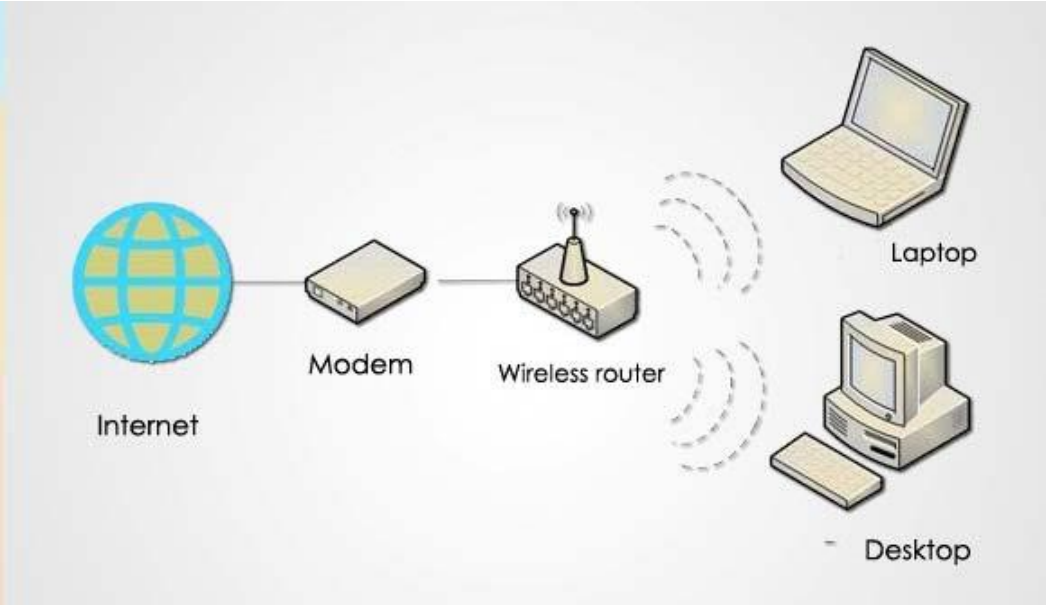
- Infrared is a media transmission system that transmits data signals through light emitting diodes (LEDs) or Lasers. Infrared is an electromagnetic energy at a wavelength which is longer than that of the red light. The information cannot be travelled through obstacles in an infrared system, but can be inhibited by light. One type of infrared is the point to point system in which transmission is possible between two points limited to a range and line of sight.
- The signal frequency to transmit in a point to point system is 100 GHz to 1,000 terahertz (THz), and the speed ranges from 100 Kbps to 16 Mbps. Another method of transmission of infrared includes the broadcast system – and, in this method, a reflective material or a transmission unit

amplifies and retransmits a data signal to several other units. The normal frequency of an infrared broadcast system is 100 GHz to 1,000 THz with a limited speed of 1 Mbps.

TYPES OF WIRELESS NETWORKS

WLANS: Wireless Local Area Networks

It allows users in a local area, such as a university campus or library, to form a network or gain access to the internet. A temporary network can be formed by a limited number of users without the need of an access point; given that they do not need access to network resources.



Two standards bodies are primarily responsible for implementing WLANs:

1)IEEE 802.11

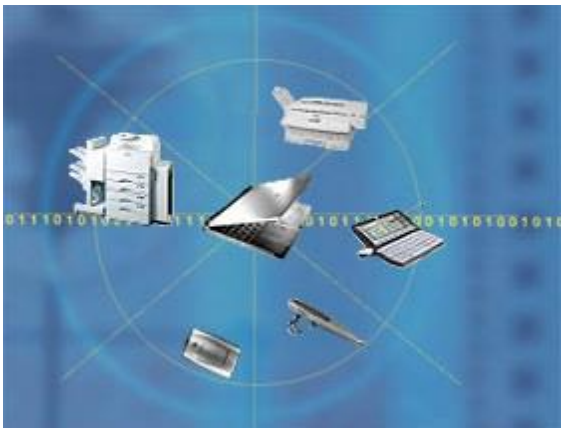
It's a non-profit organization that provides actions to coordinate produce and promote data networking standards. It defines the mechanical process of how WLANs are implemented in the 802.11 standards so that vendors can create compatible products. It specifies security association management and key management as well as access control, data confidentiality and data integrity.

2)*The Wi-Fi Alliance*

It's basically certifies companies by ensuring that their products follow the 802.11 standards, thus allowing customers to buy WLAN products from different vendors without having to be concerned about any compatibility issues.

B. WPANS: Wireless Personal Area Networks

The two current technologies for wireless personal area networks are Infra Red (IR) and Bluetooth. However, IR requires a direct line of site and the range is less. WPANs are used to convey information over short distances (about 30 feet) among a private, intimate group of participant devices. Unlike a wireless local area network (WLAN), a connection made through a WPAN involves little or no infrastructure or direct connectivity to the world outside the link. This allows small, power-efficient, inexpensive solutions to be implemented for a wide range of devices.



Three standards bodies are primarily responsible for implementing WPANs:

1) Bluetooth: A widely used WPAN technology is known as Bluetooth (version 1.2 or version 2.0) The IEEE 802.15.1 standard specifies the architecture and operation of Bluetooth devices, but only as far as physical layer and medium access control (MAC) layer operation is concerned (the core system architecture).Higher protocol layers and applications defined in usage profiles are standardized by the Bluetooth SIG (Special Interests Group).

2) ZigBee Alliance: ZigBee technology is simpler (and less expensive) than Bluetooth. The main objectives of LRWPANs (Low-Rate Wireless Personal Area Networks) like ZigBee are ease of installation, reliable data transfer,short-range operation, extremely low cost, and a reasonable battery life, while maintaining a simple and flexible protocol. The raw data rate will be high enough (maximum of 250 Kbit/s) to satisfy a set of simple needs such as interactive toys, but is also scalable down to the needs of sensor and automation needs (20 Kbit/s or below) using wireless communication.

WMANS: Wireless Metropolitan Area Network

This technology allows the connection of multiple networks in a metropolitan area such as different buildings in a city, which can be an alternative or backup to laying copper or fiber cabling.

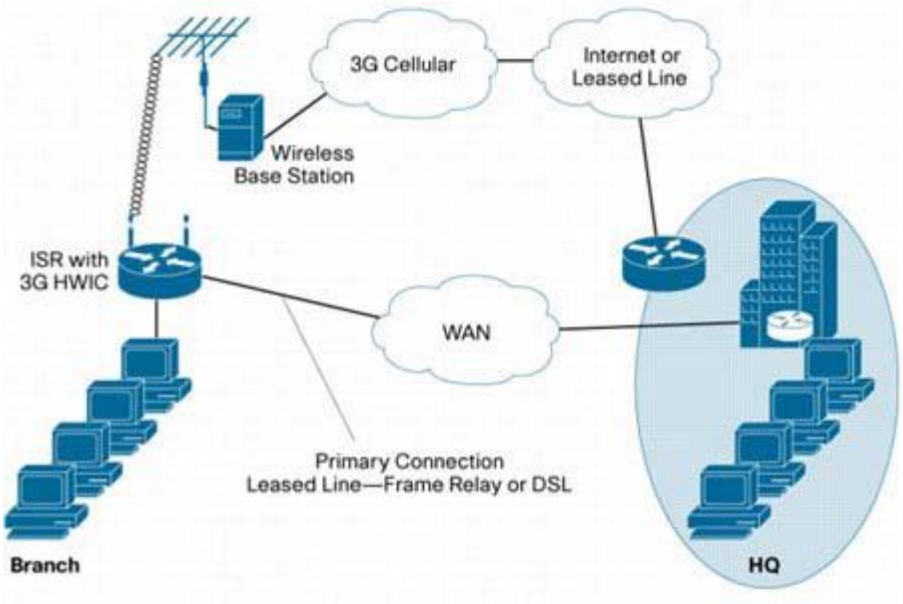
One standard body is primarily responsible for

Implementing WMANs:

1) IEEE 802.16 WiMAX: Worldwide Interoperability for Microwave Access (WiMAX) is a recent wireless broadband standard that has promised high bandwidth over long-range transmission. It is a radio frequency technology that uses licenses and unlicensed bands to provide wireless connections for non line-of-sight real implementations with speed up to 40 mbps per channel and a cell radius up to 10 kilometers for fixed and portable access situations. In the Line-Of-Sight, WiMAX can provide link distance up to 50 kilometers.

WWANS: Wireless Wide Area Networks

These types of networks can be maintained over large areas, such as cities or countries, via multiple satellite systems or antenna sites looked after by an ISP. These types of systems are referred to as 2nd Generation systems.



1) CDPD: Cellular Digital Packet Data is a technique used for transmitting small units of data, commonly referred to as packets, over the cellular network in a reliable manner.It allows to send and receive data from anywhere in the cellular coverage area at any time, quickly and efficiently. CDPD technology provides extensive, high speed, high capacity, cost effective data services to mobile users. With this technology, both voice and data can be transmitted over existing cellular channels. To effectively integrate voice and data traffic on the cellular system without degrading the level of service provided to the voice customer, the CDPD network implements a technique called channel hopping.

2)3G: Third-generation (3G) is the term for the latest generation of mobile services, which provide advanced voice communications and high-speed data connectivity, including access to the Internet, mobile data applications, and multimedia content. The International Telecommunications Union (ITU), working with industry-standards groups from around the world, has defined the technical requirements and standards as well as the spectrum for 3G systems under the International Mobile Telecommunications-2000 (IMT-2000) program.

Wireless network devices

Wireless Router

Wireless routers accepts an incoming Internet connection and sends the data as RF signals to other wireless devices that are near to the router. A network set up with a wireless router is called as a Wireless Local Area Network (WLAN).Many routers have built-in security features such as firewalls that help protect devices connected to the router against malicious data such as computer viruses.

Wireless Router

A wireless router is used in many houses to connect their computers to the Internet.

Wireless Adapters

Wireless adapters are hardware devices that are installed inside computers which enables wireless connectivity. If a computer does not have a wireless adapter, it will not be able to connect to a router in

order to access the Internet. Some computers have wireless adapters built directly into the motherboard, while it is also possible to install stand-alone wireless adapters to add wireless capability to a computer that doesn't have a built-in facility.

Wireless Repeater

A wireless repeater is a wireless networking device that is used to extend the range of a wireless router. A repeater receives wireless signals and amplifies the strength of the signals, and then re-emits them. The strength of the signal can be increased by placing a repeater between the router and the computer connected to the router.

Wireless Repeater

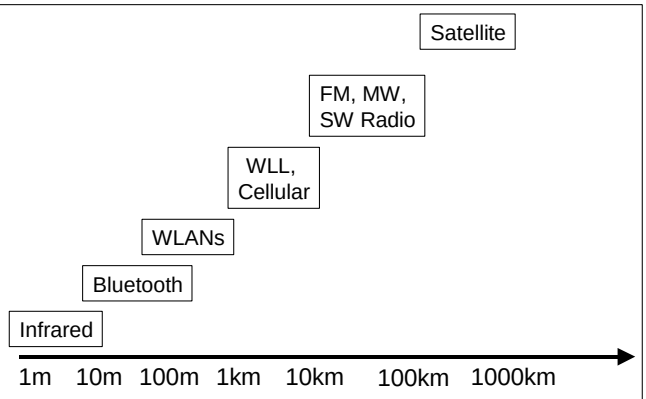


Fig. 4. Range of Wireless Systems

INTRODUCTION TO CELLULAR SYSTEM

A cellular system is a radio network made up of a number of radio cell (or just cells) each served by at least one fixed-location transceiver known as a cell site or base station. These cells cover different land areas to provide radio coverage over a wider area than the area of one cell, so that a variable number of portable transceivers can be used in any one cell and moved through more than one cell during transmission.

Cellular systems offer a number of advantages including but not limited to the following over alternative solutions:

Increased capacity
Reduced power usage
Larger coverage area
Reduced interference from other signals

Basic Cellular System

A basic cellular system is made up of three parts:

i. A mobile unit
ii. A cell site
iii. Mobile Telephone Switching Office (MTSO) and with connections to link the three subsystems:

Mobile Unit: A mobile telephone unit contains a control unit, a transceiver and an antenna system

Cell Site: The cell site provides interface between the MTSO and the mobile unit. It has control unit, radio cabinets, antennas, a power plant and data terminals.

MTSO: The switching office, the central coordinating element for all cell sites, contains the cellular processor and cellular switch. It interfaces with telephone company zone offices; controls call processing and handle billing activities.

Connections: As a matter of fact, the radio and high speed data links connect the three subsystems. Each mobile unit can only use one channel at a time for its communication link.

However, the channel is not fixed i.e. it can be any one in the entire band assigned by the serving area, with each site having multi-channel capabilities that can connect simultaneously to many mobile units.

In addition to the above:

The MTSO is the heart of the cellular mobile system. Its processor provides central coordination and cellular administration.

The cellular switch, which can be either analog or digital, switches calls to connect mobile subscribers to other mobile subscribers and to the nationwide telephone network. It uses voice trucks similar to telephone company interoffice voice trucks. It also contains data links providing supervision links between the processor and the switch and between the cell sites and the processor.

The radio link carries the voice and signal between the mobile unit and the cell site.

The high-speed data links cannot be transmitted over the standard telephone trucks and therefore must use either microwave links or T-carriers (wire lines).

Microwave radio links or T-carriers carry both voice and data between the cell site and the MTSO.

CELLULAR WIRELESS: EVOLUTION OF GSM TECHNOLOGY

Introduction

The explosive growth of Global System for Mobile (GSM) Communication services over the last two decades has changed mobile communications from a niche market to a fundamental constituent of the global telecommunication markets. GSM is a digital wireless technology standard based on the notion that users want to communicate wirelessly without limitations created by network or national borders. In

a short period of time, GSM has become a global phenomenon. The explanation for its success is the cooperation and coordination of technical and operational evolution that has created a virtuous circle of growth built on three principles: interoperability based on open platforms, roaming and economies of scale.

B. Global System for Mobile Communications (GSM)

Introduction

The European Telecommunications Standards Institute (ETSI) defines GSM as an *international cellular telephony standard*. Through this standard, it is relatively easy today for a person with a GSM phone to travel from Europe to the Middle East, Australia and Africa and be able to use a single phone.

> *GSM became popular very quickly because it provided* ***improved speech quality*** *and, through a uniform international standard, made it possible to use a* ***single mobile unit around the world****.*

Recall that the wireless personal communications industry has moved from the traditional analog technology such as FM (FDMA) to digital technologies such as TDMA and CDMA. Four reasons account for this:

- ***Quality of service*** (*voice clarity*) – especially in a harsh environment such as in mobile radio

- ***Increased capacity*** – compression techniques can reduce the number of bits needed to represent speech

- ***Privacy/Security*** – a digital communication system provides a ready platform where encryption techniques may be used to safeguard information transmitted over the air.

- ***Enabling technology and economics*** – in recent years, the economics of electronic technologies (such as digital signal processing and integrated circuits) has made them feasible for use in commercial application on a mass scale.

GSM is a digital standard that uses TDMA as a multiple access technology, and its benefits include:

- *Support for international roaming;*
- *Excellent speech quality;*
- *Wide range of services (sms, fax, videotext, etc);*
- *Extensive security features;*
- *Interworking (with PSTN, etc)*

The principal components of a digital communications system are shown:

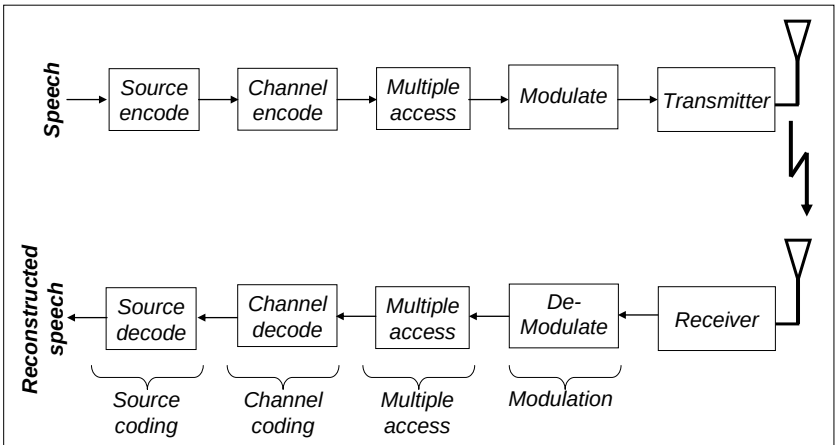


Fig. 2. Principal Components of a digital communications system

The basic architecture for the 2G wireless systems is shown in Fig. 7:

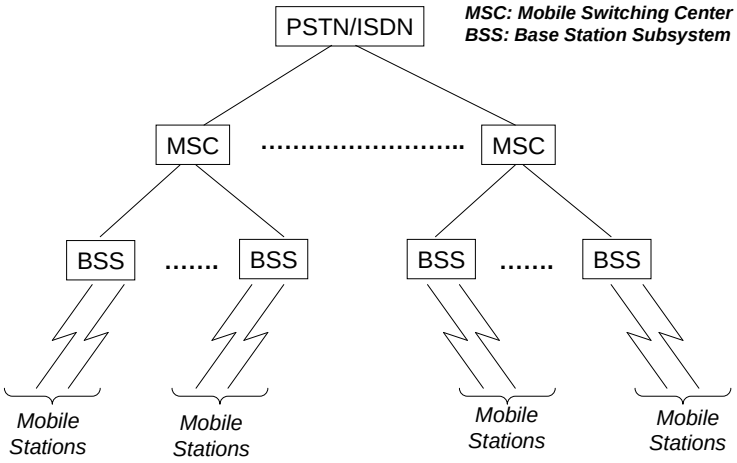


Fig. 7. Basic Architecture of 2G Cellular Mobile Systems

The GSM network architecture is shown in Fig, 7 where:

- MS – Mobile Station (vehicle mounted units, portable units, handheld units)
- BTS – Base Transceiver Station;
- BSS – Base Station Subsystem;
- BSC – Base Station Controller;
- HLR (Home Location Register); VLR (Visitor Location Register); EIR (Equipment Identity Register); AUG (Authentication Center)
- MSC – Mobile Switching Center.

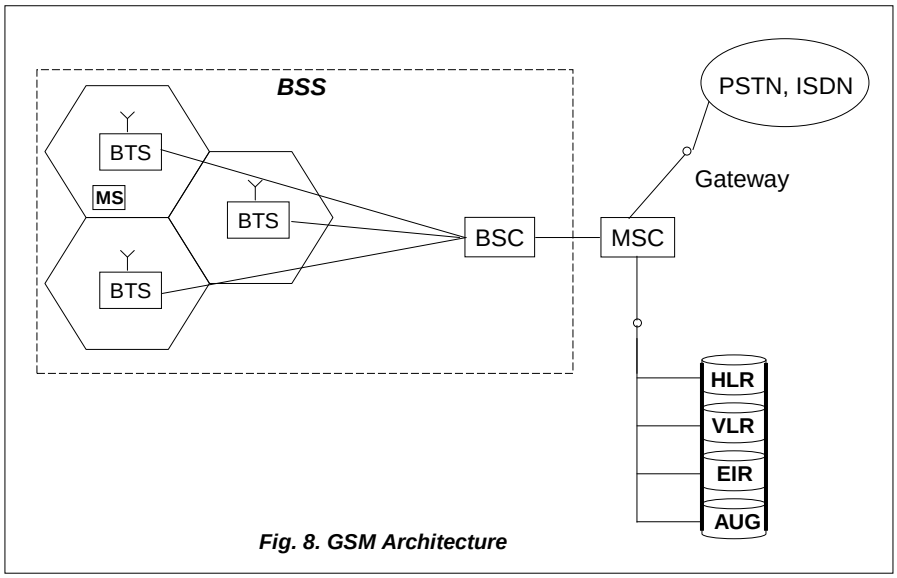


Fig. 8. GSM Architecture

The wireless system is connected to the wired telephony network – the Public Switched Telephone Network, PSTN – through mobile switching centers (MSC). Each MSC supports several base station subsystems (BSS). Each BSS includes a base station controller (BSC) that supports several base station transceiver stations (BTS). The mobile stations (MSs) are connected to the cellular system through a BTS with an air interface define by the GSM standard.

The GSM architecture can be divided into three subsystems:

- Mobile Station (MS);
- Base Station Subsystem (BSS);
- Network Subsystem (NSS)

(a). Mobile Station (MS)

Communicates across an *air interface* with a BTS in the same cell in which the mobile unit is located. The mobile equipment includes:

- ***Radio receiver****;* ***Digital signal processors****;* and ***Subscriber Identify Module*** *(SIM)*

The SIM card stores the following information:

- *Subscriber identification number*;
- *The networks the subscriber is authorized to use;*
- *Encryption keys;*
- *Other information specific to subscriber*

(b). Base Station Subsystem (BSS)

This unit consists of a base station controller (BSC) and one or more base transceiver stations (BTS). The BSC reserves radio frequencies, manages the handoff of a mobile unit from one cell to another within the BSS, etc.

The BSS is realized as a network of neighbouring radio cells together providing a complete coverage of the service area. Each cell has a Base Transceiver Station (BTS) operating on a set of radio channels, which are different from those used in neighbouring cells to avoid interference

(c). Network Subsystem (NSS)

The NSS provides the following:

- *The link between the cellular network and the PSTNs*;

- *It controls handoff between cells in different BSSs;*
- *Authenticates users and validates their accounts;*
- *Includes functions for worldwide roaming of mobile users*.

The central element of the NS is the MSC, but several databases are also part of the NSS:

(i). Home Location Register (HLR)
Information about the location of the Mobile Station (MS) is stored here (i.e., the actual MSC the MS is associated with). This information may change while the MS moves. The HLR also contains the services associated with each MS.

(ii). Visitor Location Register (HLR)
The VLR contains information about all MSs in this MSC area that are switched on. As soon as an MS roams into a new MSC area, the VLR connected to that MSC will request data about the MS from the HLR.

(iii). Equipment Identify Register (EIR)
Each mobile station has its own mobile Equipment Identity (EI). The EIR is a database that contains a list of all valid mobile station equipment within the network. The EIR has three databases:

- White list – for all known, good EIs;
- Black list – for bad or stolen handsets;
- Gray list – handsets/EIs that are uncertain

(iv). Authentication Center
A protected database that holds a copy of the secret key stored in each subscriber's SIM card, which is used for *authentication* **and** *encryption* **over the radio**

Generation of Cellular Wireless

The generations of cellular wireless are:
First Generation (1G)
Second Generation (2G)
Third Generation (3G)
Fourth Generation (4G)

Table 1. Different generations of commercial cellular networks.

Generation/ year	Key features and access capabilities	Reliability, QoS, and performance	Access protocols
1G/1982	– Uses analog signals, primarily for voice communications but some support for low rate data – Supports access to and roaming across a single type of analog wireless networks	– Affected by limited coverage – Rudimentary QoS – Performance didn't meet expectations	Frequency Division Multiple Access (FDMA), where each user is assigned a channel for the duration of the call
2G/1991	– Uses digital technology for secure voice and data – Supports access to and roaming across a single type of digital wireless network – Offers access to 1G networks	– Improved reliability with backup access to 1G networks – Better QoS – Performance didn't meet expectations	Time Division Multiple Access (TDMA), where a user time-shares a channel, or Code Division Multiple Access (CDMA), where a user is assigned a code for the duration of the call
3G/2000	– Supports multimedia content – Supports global roaming across a single type of wireless network (such as a cellular network), at speeds of 384 Kbps to several Mbps – Offers access to 2G networks	– Improved reliability with backup access to 2G (sometimes 1G) networks – Improved QoS – Performance is meeting expectation	TDMA and several variations of CDMA, including Wideband CDMA
4G/2010	– Supports global roaming across multiple high-bandwidth wireless networks (50 Mbps or even higher) – Offers IP interoperability for seamless access to the mobile Internet – Offers access to 2G and 3G networks	– Improved reliability with backup access to 3G (sometimes 2G) networks – Highly improved QoS – Performance is somewhat meeting expectations	Orthogonal Frequency Division Multiple Access with multiple antennas (single and multicarrier)

Mobile Phone Networks

The most common example of a cellular network is a mobile phone (cell phone) network. A mobile phone is a portable telephone which receives or makes calls through a cell site (base station), or transmitting tower. Radio waves are used to transfer signals to and from the cell phone.

Large geographic areas (representing the coverage range of a service provider) may be split into smaller cells to avoid line-of-sight signal loss and the large number of active phones in an area. In cities, each cell site has a range of up to approximately ½ mile, while in rural areas; the range is approximately 5 miles. Many times in clear open areas, a user may receive signals from a cell site 25 miles away. All of the cell sites are connected to cellular telephone exchanges "switches", which connect to a public telephone network or to another switch of the cellular company.

As the phone user moves from one cell area to another cell, the switch automatically commands the handset and a cell site with a stronger signal (reported by each handset) to switch to a new radio channel (frequency). When the handset responds through the new cell site, the exchange switches the connection to the new cell site.

With CDMA, multiple CDMA handsets share a specific radio channel. The signals are separated by using a pseudonoise code (PN code) specific to each phone. As the user moves from one cell to another, the handset sets up radio links with multiple cell sites (or sectors of the same site) simultaneously. This is known as "soft handoff" because, unlike with traditional cellular technology, there is no one defined point where the phone switches to the new cell.

Modern mobile phone networks use cells because radio frequencies are limited and shared resource. Cell-sites and handsets change frequency under computer control and use low power transmitters so that a limited number of radio frequencies can be simultaneously used by many callers with less interference.

Since almost all mobile phones use cellular technology, including GSM, CDMA, and AMPS (analog), the term "cell phone" is used interchangeably with "mobile phone". However, satellite phones are mobile phones that do not communicate directly with a ground-based cellular tower, but may do so indirectly by way of a satellite.

Cellular Concept

Cellular telephone systems must accommodate a large number of users over a large geographic area with limited frequency spectrum, i.e., with limited number of channels. If a single transmitter/ receiver is used with only a single base station, then sufficient amount of power may not be present at a huge distance from the BS.

For a large geographic coverage area, a high powered transmitter therefore has to be used. But a high power radio transmitter causes harm to environment. Mobile communication thus calls for replacing the high power transmitters by low power transmitters by dividing the coverage area into small segments, called cells. Each cell uses a certain number of the available channels and a group of adjacent cells together use all the available channels. Such a group is called a cluster. This cluster can repeat itself and hence the same set of channels can be used again and again. Each cell has a low power transmitter with a coverage area equal to the area of the cell. This technique of substituting a single high powered transmitter by several low powered transmitters to support many users is the backbone of the cellular concept

Operational Channels

In each cell, there are four types of channels that take active part during a mobile call. These are:
_ Forward Voice Channel (FVC): This channel is used for the voice transmission from the BS to the MS.
_ Reverse Voice Channel (RVC): This is used for the voice transmission from the MS to the BS._ Forward Control Channel (FCC): Control channels are generally used for controlling the activity of the call, i.e., they are used for setting up calls and to divert the call to unused voice channels. Hence these are also called setup channels. These channels transmit and receive call initiation and service request messages. The FCC is used for control signaling purpose from the BS to MS.
_ Reverse Control Channel (RCC): This is used for the call control purpose from the MS to the BS. Control channels are usually monitored by mobiles.

Making a Call

When a mobile is idle, i.e., it is not experiencing the process of a call, then it searches all the FCCs to determine the one with the highest signal strength. The mobile then monitors this particular FCC. However, when the signal strength falls below a particular threshold that is insufficient for a call to take place, the mobile again searches all the FCCs for the one with the highest signal strength. For a particular country or continent, the control channels will be the same. So all mobiles in that country or continent will search among the same set of control channels. However, when a mobile moves to a different country or continent, then the control channels for that particular location will be different and hence the mobile will not work. Each mobile has a mobile identification number (MIN). When a user wants to make a call, he sends a call request to the MSC on the reverse control channel. He also sends the MIN of the person to whom the call has to be made. The MSC then sends this MIN to all the base stations. The base station transmits this MIN and all the mobiles within the coverage area of that base station receive the MIN and match it with their own. If the MIN matches with a particular MS, that mobile sends an acknowledgment to the BS. The BS then informs the MSC that the mobile is within its coverage area. The MSC then instructs the base station to access specific unused voice channel pair. The base station then sends a message to the mobile to move to the particular channels and it also sends a signal to the mobile for ringing.

In order to maintain the quality of the call, the MSC adjusts the transmitted power of the mobile which is usually expressed in dB or dBm. When a mobile moves from the coverage area of one base station to the coverage area of another base station i.e., from one cell to another cell, then the signal strength of the initial base station may not be sufficient to continue the call in progress. So the call has to be transferred

to the other base station. This is called handoff. In such cases, in order In such cases, in order to maintain the call, the MSC transfers the call to one of the unused voice channels of the new base station or it transfers the control of the current voice channels to the new base station.

Concept of Cell Cluster

When devising the infrastructure technology of a cellular system, the interference between adjacent channels is reduced by allocating different frequency bands or channels to adjacent cells so that their coverage can overlap slightly without causing interference. In this way cells can be grouped together in what is termed a cluster. A group of cells that use different set of frequencies in each cell is called a Cell Cluster. Let N be the cluster size in terms of the number of cells within it and K be the total number of available channels without frequency reuse. The N cells in the cluster would then utilise all K available channels. In this way, each cell in the cluster contains one-Nth of the total number of available channels. In this sense, N is also referred to as the frequency reuse factor of the cellular system.

Seven is a convenient number of cells that can be in a cluster, but there are number of conflicting requirements that need to be balanced when choosing the number of cells in a cluster for a cellular system:

Limiting interference levels Number of channels that can be allocated to each cell site It is necessary to limit the interference between cells having the same frequency. The topology of the cell configuration has a large impact on this. The larger the number of cells in the cluster, the greater the distance between cells sharing the same frequencies. In the ideal world it might be good to choose a large number of cells to be in each cluster.

Unfortunately there are only a limited number of channels available.

This means that the larger the number of cells in a cluster, the smaller the number available to each cell, and this reduces the capacity. This means that there is a balance that needs to be made between the number of cells in a cluster, and the interference levels and the capacity that is required.

(a) Capacity Increase by Frequency Reuse Concept

Let us consider that each is allocated J channels ($J \leq K$). If the K channels are divided among the N cells into unique and disjoint channel groups, each with J channels, then we have

$K = J N$

N cells in a cluster use the complete set of available frequencies while K is the total number of available channel. By decreasing the cluster size N, it is possible to increase the capacity per cell. The cluster can be replicated many times to form the entire cellular communication system.

Let M be the number of times the cluster is replicated and C be the total number of channels used in the entire cellular system with frequency reuse. C is then the system capacity and is given by

$C = M J N$

If M is increased, the system capacity C will increase.

EXERCISE 4

Given a cellular system which has a total of 1001 radio channels available for handling traffic and given the area of a cell is 6 km2 and the area of the entire system is 2100 km2.

i. Calculate the system capacity if the cluster size is 7.

ii. How many times would the cluster of size 4 have to be replicated

in order to approximately cover the entire cellular area?

iii. Calculate the system capacity if the cluster size is 4.

iv. Does decreasing the cluster size increase the system capacity?

Explain

Solution: Given that

The total number of available channels K = 1001.
Cluster size N = 7.
Area of cell Acell = 6 km2
Area of cellular system $A_{sys} = 2100\ km^2$

(i) The number of channels per cell is given by $J \quad \frac{K}{N}$

Then we have $J \quad \frac{1001}{7} \quad 143\ channels / cell$

Also, we know that the coverage area of a cluster is given by
$A_{cluster} \quad N \quad A_{cell} \quad 7 \quad 6 \quad 42 km^2$

The number of times that the cluster has to be replicated to cover the entire system will be $M \quad \frac{A_{sys}}{A_{cluster}}$ or $M \quad \frac{2100}{42} \quad 50$

Hence, the system capacity C will be

C = M J N = 50 x 143 x 7 = 50,050 channels.

(ii) For N = 4, $A_{cluster}$ = 4 x 6 = 24 km^2

Hence $M \quad \frac{A_{sys}}{A_{cluster}} \quad \frac{2100}{24} \quad 87.5 \quad 87.$

(iii) With N = 4, $J \quad \frac{1001}{4} \quad 250 \; channels / cell.$

The system capacity is then

C = 87 x 250 x 4 = 87,000 channels.

From (i) and (iii), it is obvious that a decrease in N from 7 to 4 is accompanied by an increase in M from 50 to 87, and the system capacity is increased from 50,050 channels to 87,000 channels.

Hence, decreasing the cluster size does increase the system capacity.

Rule to Determine the Nearest Co-channel Neighbours

The following two-step rule can be used to determine the location of the nearest co-channel cell:

Step 1: Move i cells along any chain of hexagons;

Step 2: Turn 60 degrees counter clockwise and move j cells. Figure 4.5 shows the method of locating co-channel cells in a cellular system using the preceding rule for i = 3 and j= 2, where the co-channel cells are the shaded cells.

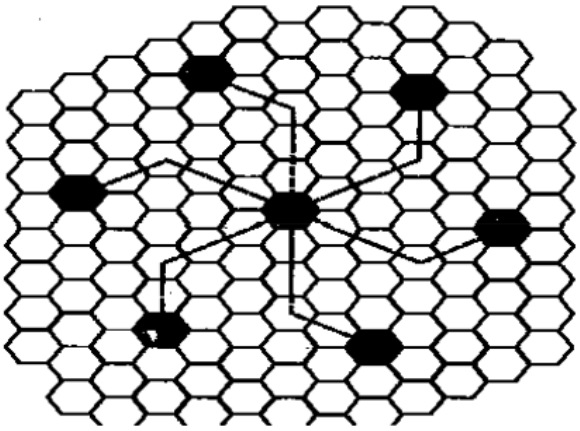


Fig. 4.5 Locating Co-Channel Cells in a Cellular System

Figure 4.6 illustrates the cluster concept and frequency reuse in a cellular network where cells with the same number use the same set of

frequencies. These are co-channel cells that must be separated by a distance such that the co-channel interference is below a prescribed threshold. The parameters i and j measure the number of nearest neighbours between co-channel cells; the cluster size, N, is related to I and j by the following expression:

N = i2 + ij + j2

For example, if i = 1 and j = 2, then, N = 7. With a cluster size N = 7, the frequency reuse factor is seven since each cell contains one-seventh of the total number of available channels.